PropertyTrust

Property Insurance Management System

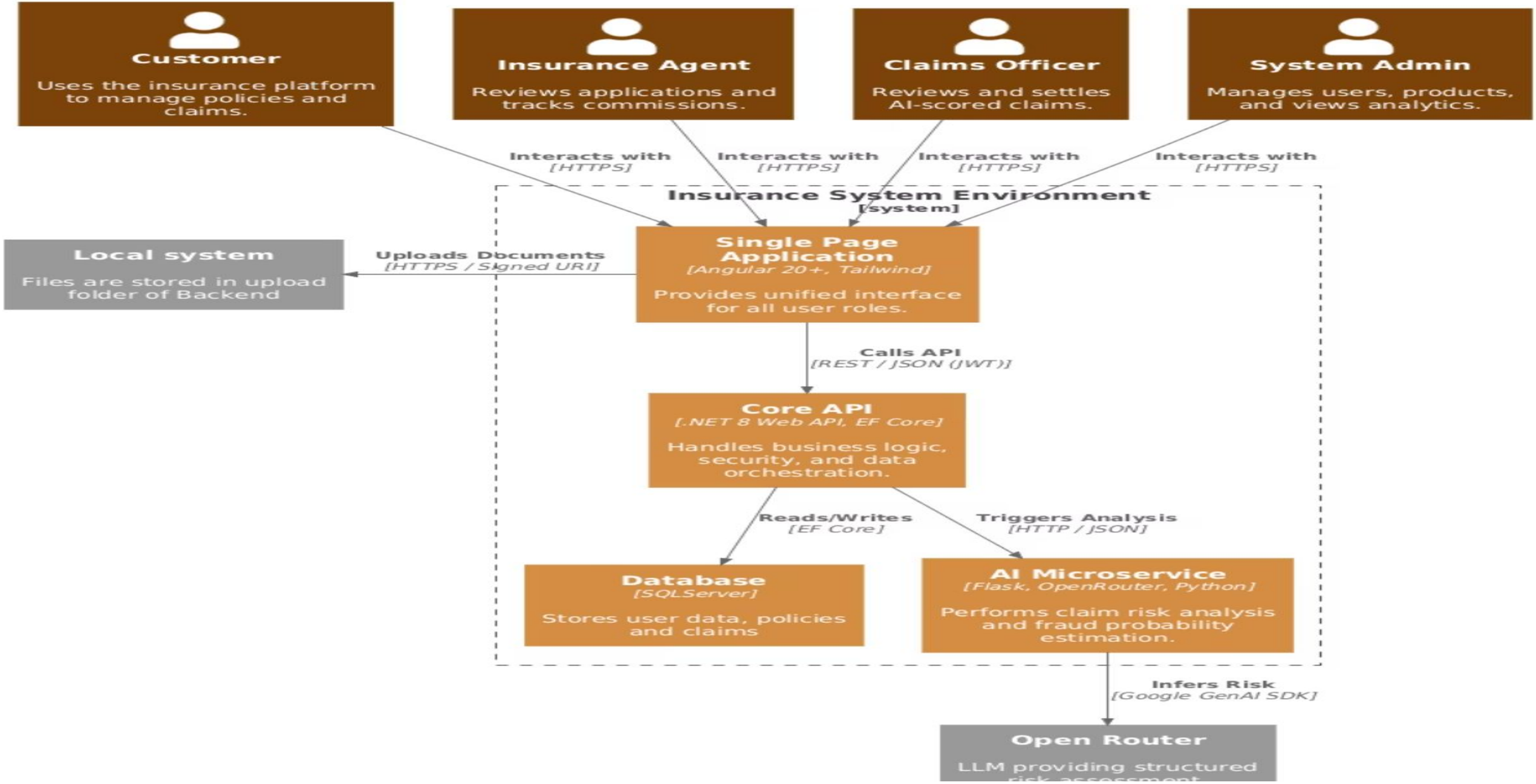
Author: Gurumetkal Bharath Simha Reddy

Guidance: Ranjan Bhatnagar

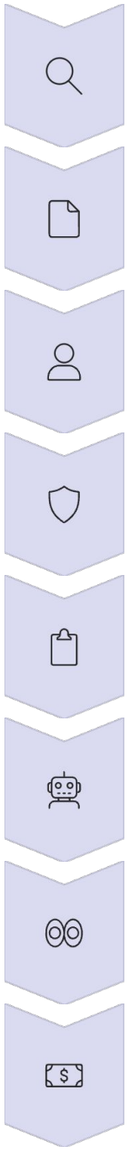
Abstract

This project presents an AI-Driven Property Insurance Management System that digitizes the complete insurance lifecycle from product discovery to claim settlement. The platform integrates Angular 20+ frontend, .NET Web API backend, and a Flask-based AI microservice to enable intelligent risk scoring, fraud probability estimation, commission automation, and role-based operational workflows. By incorporating AI-assisted decision support and risk-based prioritization, the system enhances operational efficiency, transparency, and scalability while maintaining secure, modular, and enterprise-grade architecture.

System Architecture - Property Insurance Platform



Business Workflow



Product Discovery

Application Submission

Agent Approval

Policy Activation

Claim Submission

AI Risk Scoring

Claims Review

Settlement & Commission

Role-Based Responsibilities

Customer

- Browse insurance products
- Submit insurance applications
- Pay policy premium
- View active and expired policies
- Submit claims with supporting documents
- Track claim status

Agent

- Review insurance applications
- Approve or reject applications
- Manage assigned customers
- Track earned commissions

Claims Officer

- Review submitted claims
- View AI risk score and fraud probability
- Prioritize claims based on risk
- Approve, reject, or partially settle claims

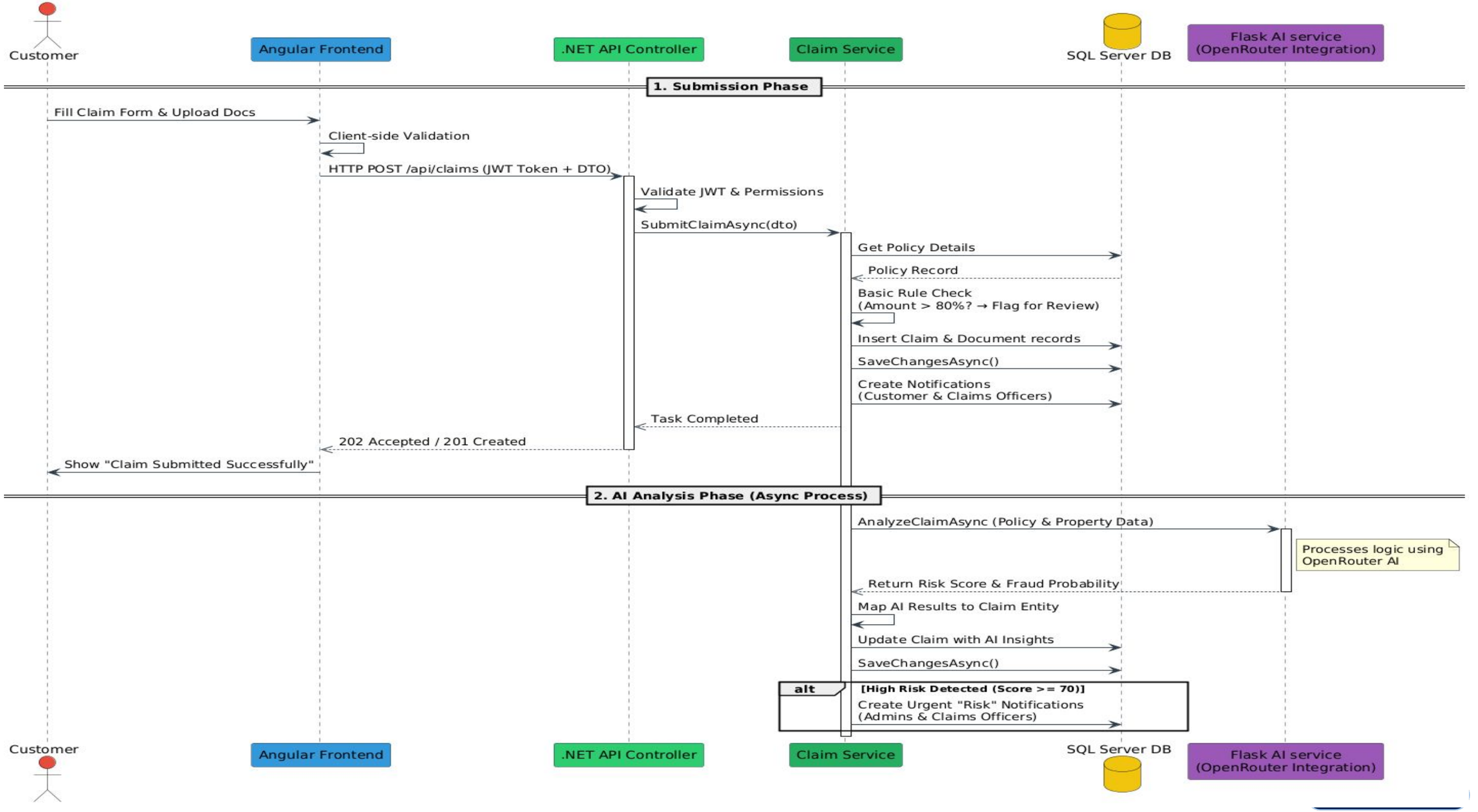
Admin

- Monitor system-wide operations
- View AI analytics dashboard
- Analyze fraud trends
- Evaluate agent performance
- Monitor high-risk claims
- Manage product configurations

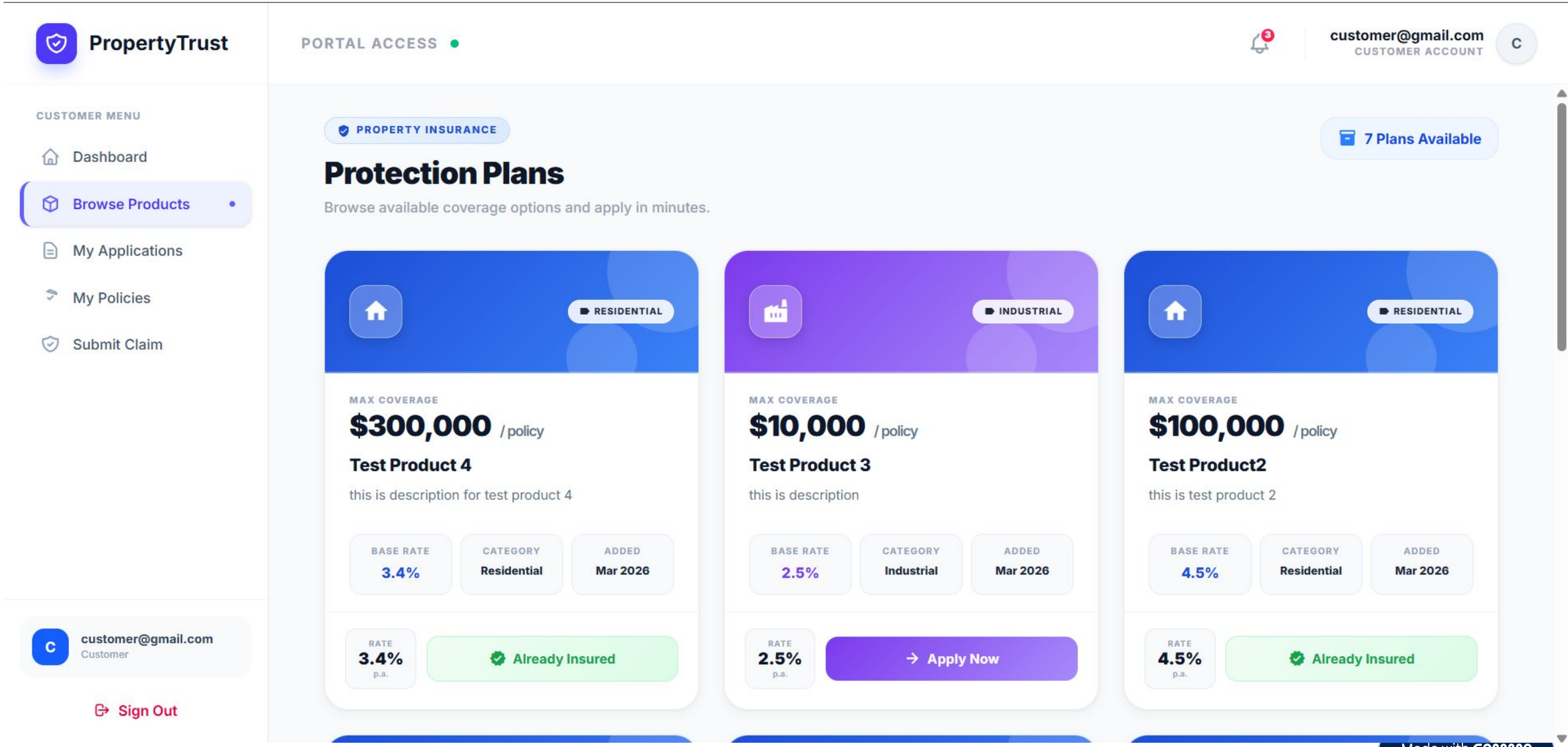
Core Entities & Descriptions

Entity	Description	Key Attributes
User	Central identity table for all actors	Name, Email, Role, IsActive
Policy Application	Digital request form for new insurance	ProductId, CustomerId, MarketValue, Status
Policy	Active insurance contract	PolicyNumber, CoverageAmt, Premium, EndDate
Claim	Incident report filed by customer	ClaimAmount, AI Risk Score, AI Fraud Prob, Status
Agent	Professionals managing customers	LicenseNumber, JoinedDate, UserId
Property	Insured asset (Home/Business)	Address, MarketValue, RiskZone, YearBuilt
Payment	Financial history of premiums	Amount, PaymentDate, PaymentStatus
Commission	Agent earnings per policy	CommissionRate, CommissionAmount, IsPaid

Made with GAMMA

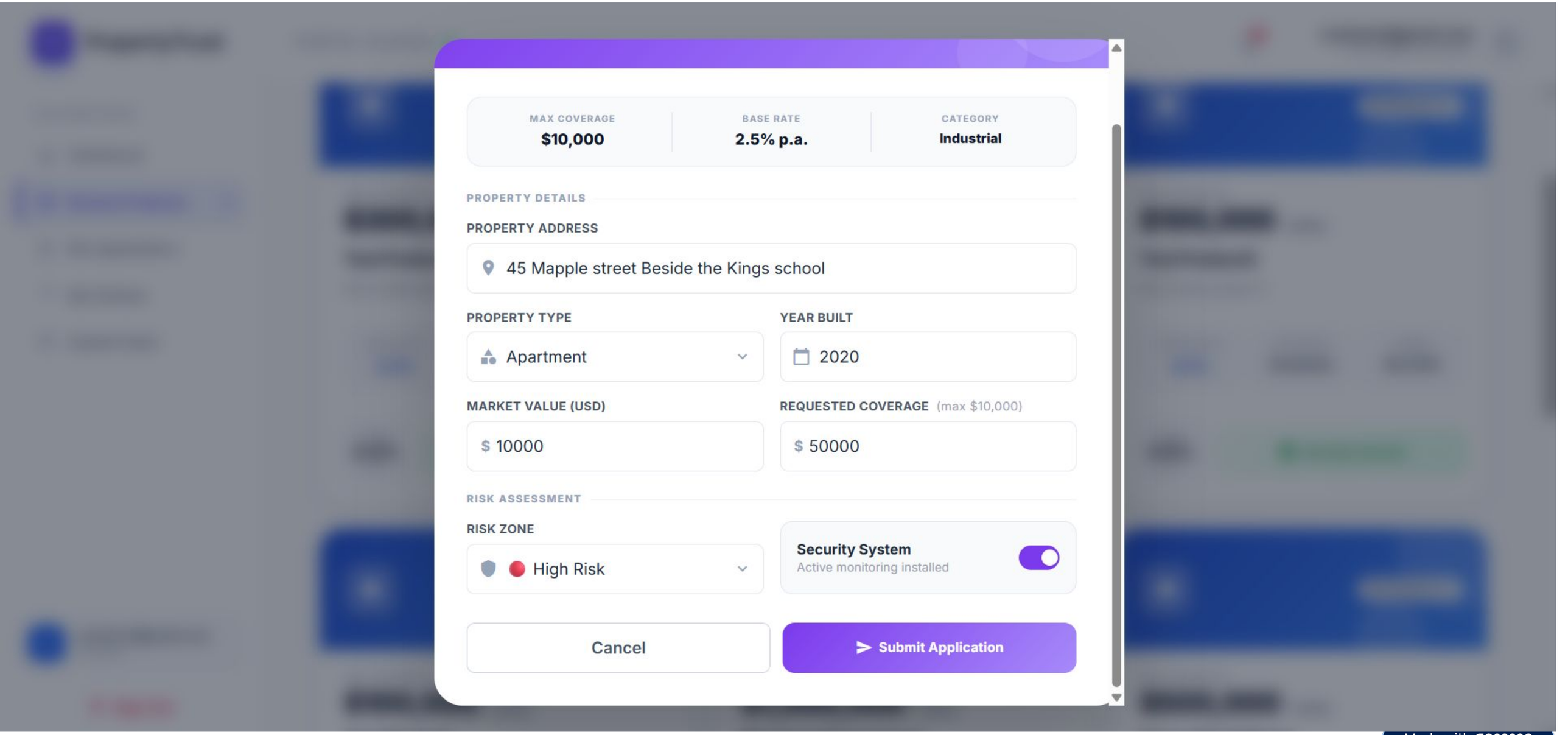

Customer
Angular Frontend
.NET API Controller
Claim Service
SQL Server DB
Flask AI service
(OpenRouter Integration)
1. Submission Phase
Fill Claim Form & Upload Docs
Client-side Validation
HTTP POST /api/claims (JWT Token + DTO)
Validate JWT & Permissions
SubmitClaimAsync(dto)
Get Policy Details
Policy Record
Basic Rule Check
(Amount > 80%? → Flag for Review)
Insert Claim & Document records
SaveChangesAsync()
Create Notifications
(Customer & Claims Officers)
Task Completed
202 Accepted / 201 Created
Show "Claim Submitted Successfully"
2. AI Analysis Phase (Async Process)
AnalyzeClaimAsync (Policy & Property Data)
Processes logic using
OpenRouter AI
Return Risk Score & Fraud Probability
Map AI Results to Claim Entity
Update Claim with AI Insights
SaveChangesAsync()
alt
[High Risk Detected (Score >= 70)]
Create Urgent "Risk" Notifications
(Admins & Claims Officers)
Customer
Angular Frontend
.NET API Controller
Claim Service
SQL Server DB
Flask AI service
(OpenRouter Integration)

Customer Browsing products



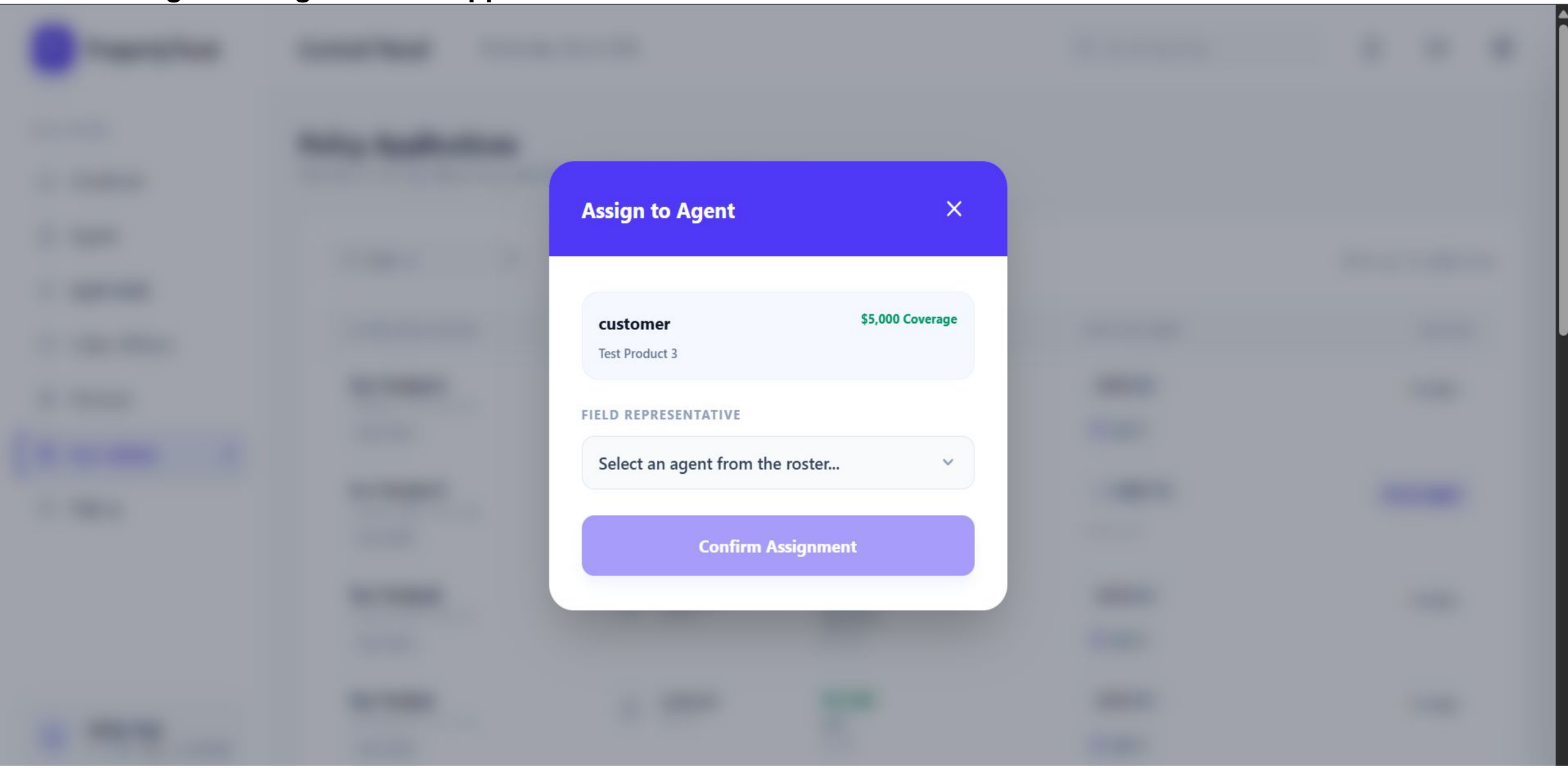
Made with GAMMA

Customer fills the Application

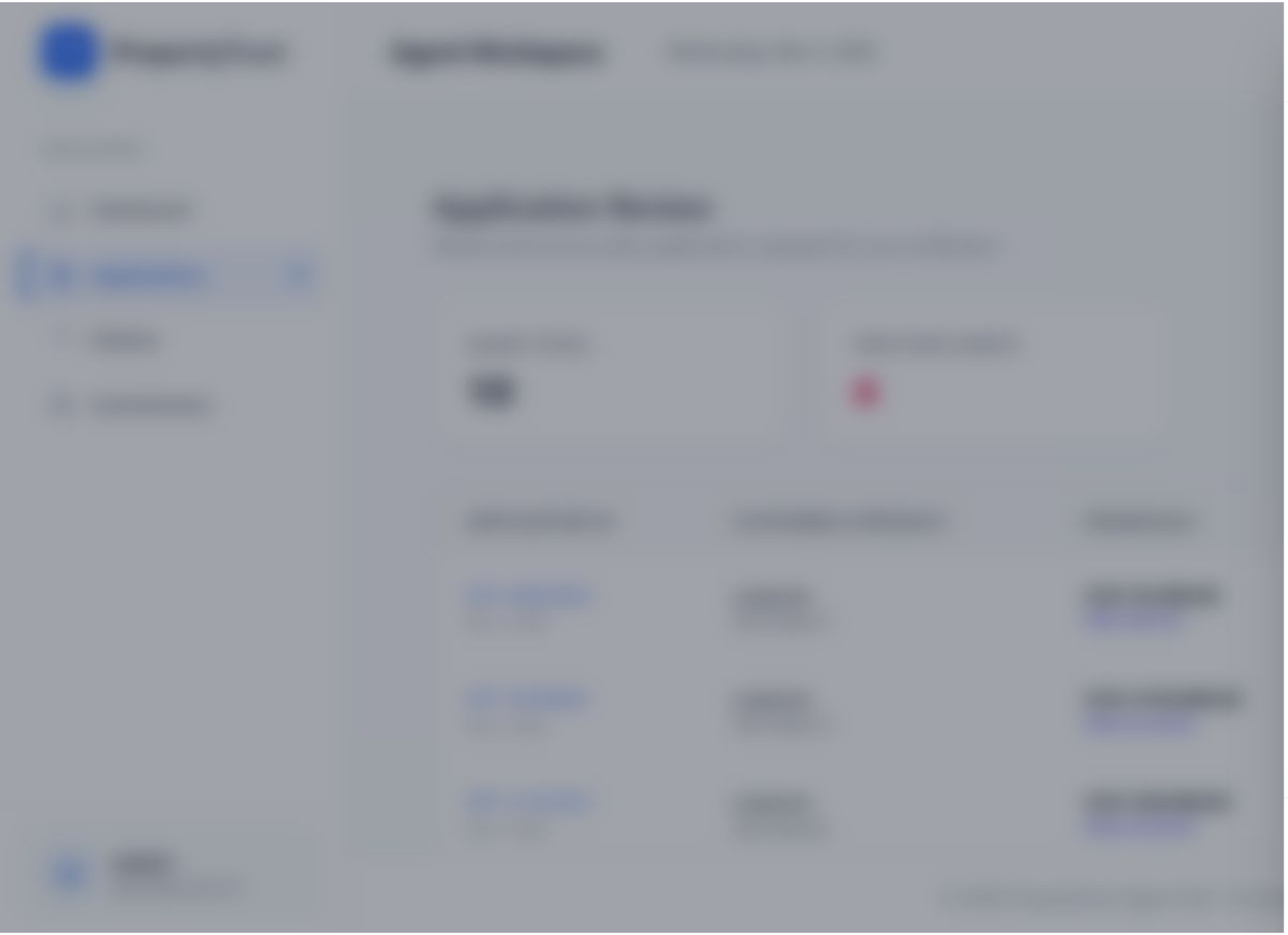


Made with GAMMA

Admin Assigns the Agent to the Application



Agent Approves/Reject Application



Application Audit

RET-ID: 85863BA9-248C-42DF-8FC6-EB6E44337DE5

PRODUCT SPECIFICATION

Selected Product
Test Product 3

Desired Coverage	Calculated Premium
$5,000.00	**$625.00**

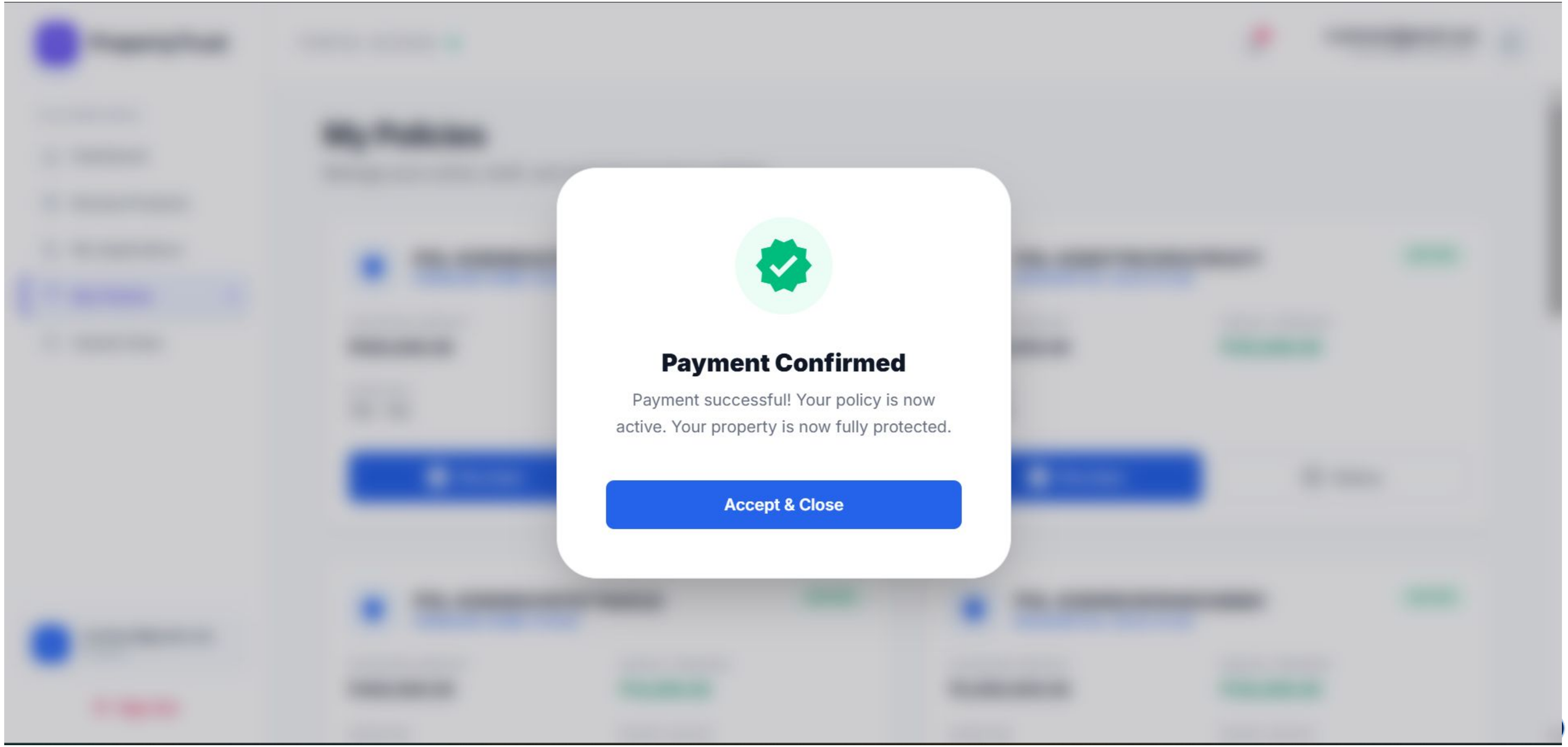
CUSTOMER CONTEXT

Applicant Name	**customer**
Submission Date	March 4, 2026
Current Status	ASSIGNEDTOAGENT

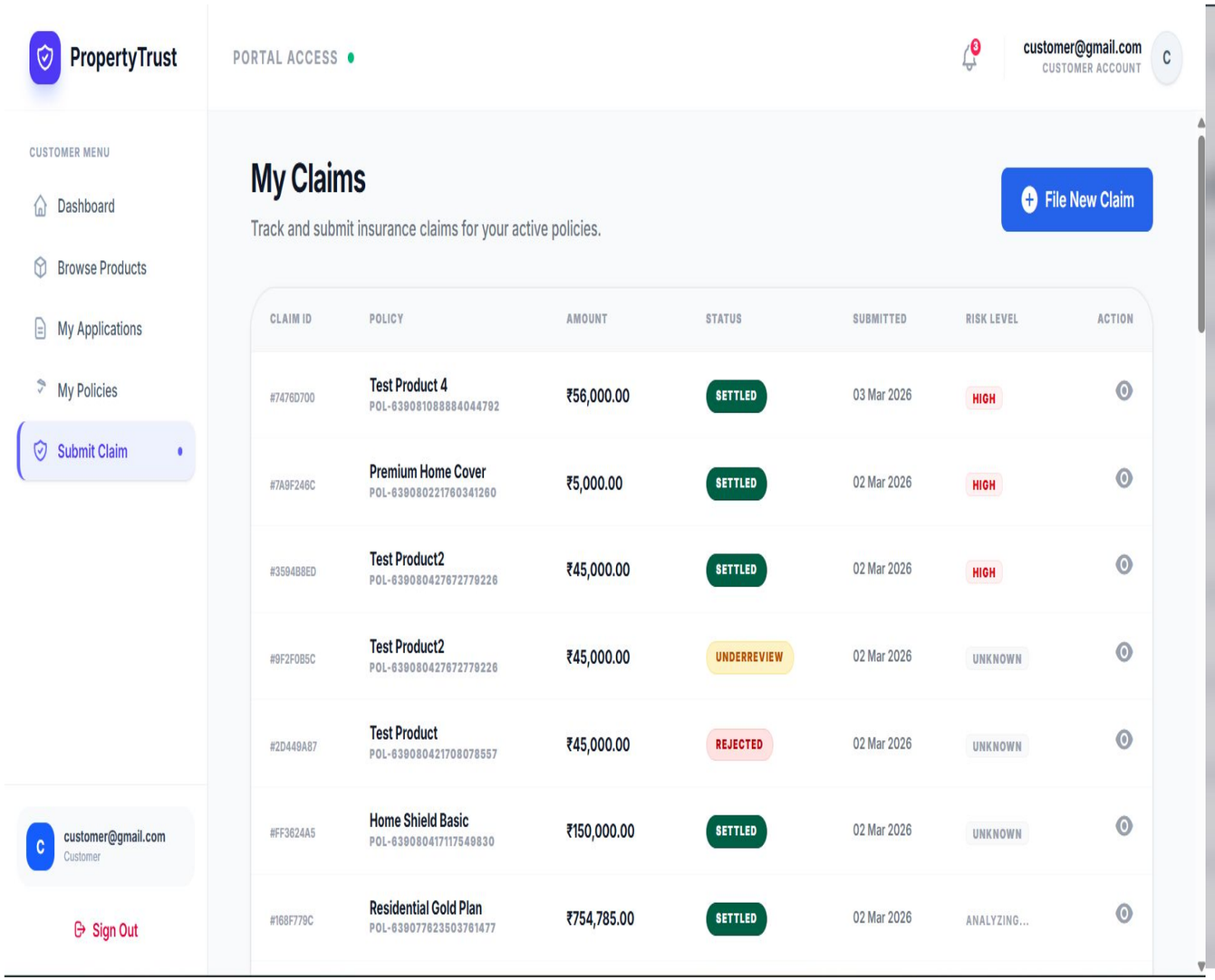
PROPERTY APPRAISAL

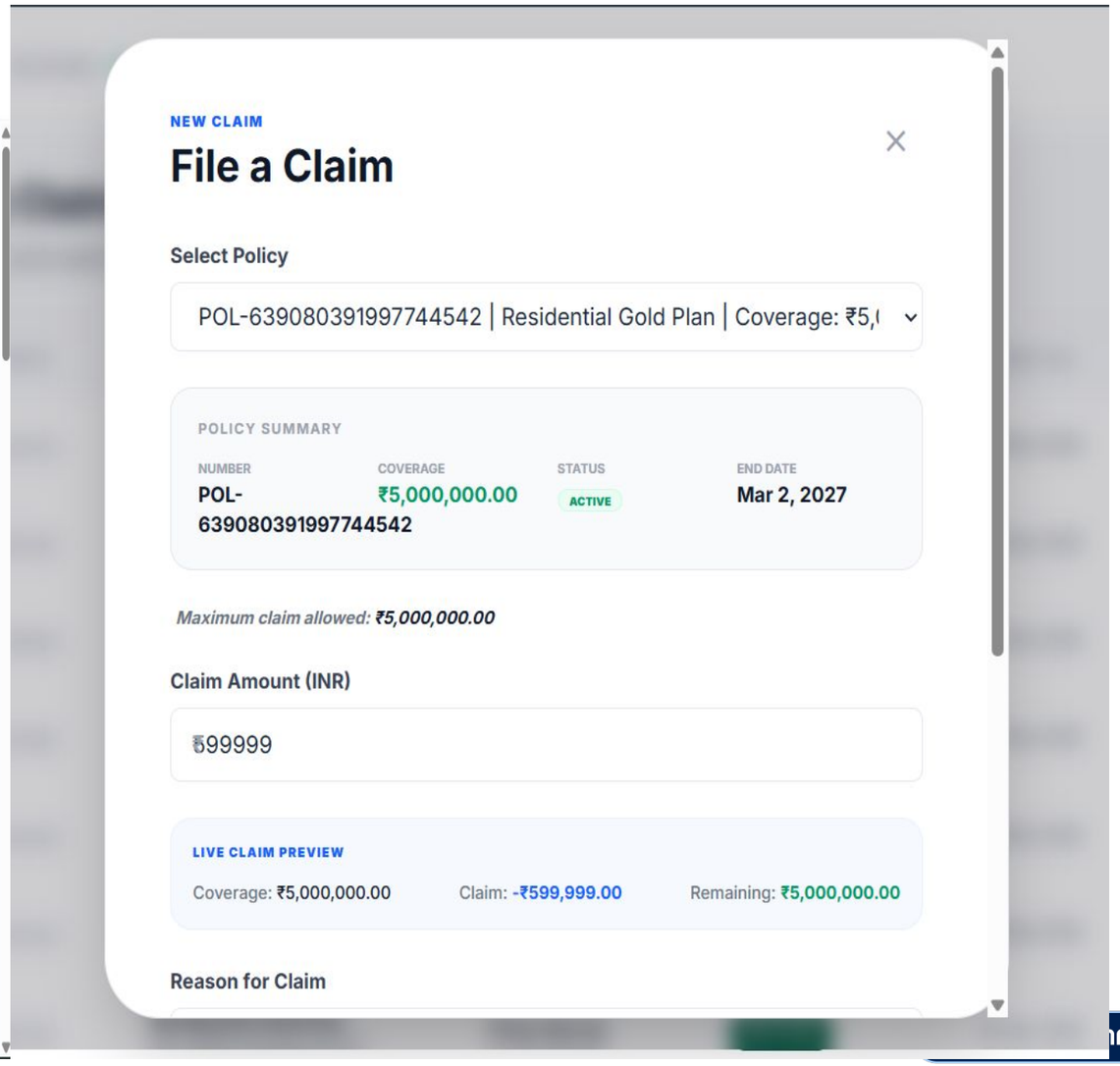
APPROVE APPLICATION | **REJECT**

Customer Makes Payment

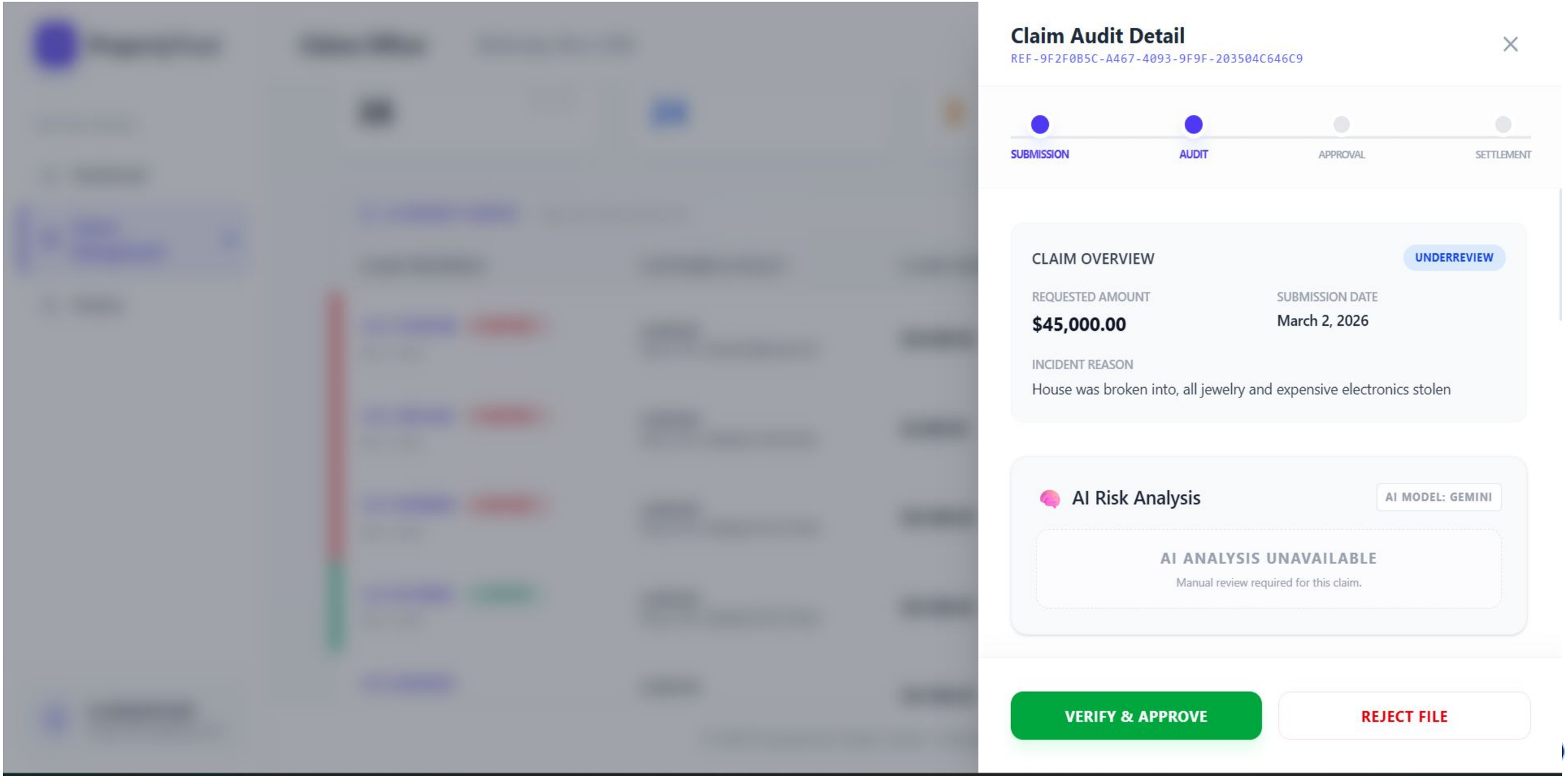


Customer Applies for a claim and Keep Track of it





Claimsoffcier verifies approves and does settlement



AI Implementation

- Structured prompt engineering
- Risk score calculation
- Fraud probability
- Heuristic fallback mechanism
- Explainable AI output

Testing Strategy

Unit testing (xUnit, Moq)

AI fallback testing

Key Differentiators

AI risk prioritization

Explainable AI model

Future Enhancements

ML-based fraud model

Renewal automation

Policy endorsement feature

Document validation using AI

Customer support using a Bot